QP Code: 25901002

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR / IMPROVEMENT / REAPPEARANCE
EXAMINATION NOVEMBER 2025
FIRST SEMESTER

Discipline Specific Core Course (DSC) – MG1DSCZGY101 – BIOLOGICAL BASIS OF BEHAVIOUR – I
(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Fill in the blanks
Answer All Questions
Each Question carries 1 mark

1. The person who treats people with emotional distress or troublesome behaviors, sometimes using drugs or other medical procedures is **Psychiatrist.** **[U]/[CO1]**
2. Operant conditioning was first described by psychologist **B. F. Skinner.** (Liberal Valuation. Give full marks if attempted) **[U]/[CO1]**
3. The voltage measured when the inside of the cell is negative relative to the outside is known as the **Resting membrane/ membrane** potential. **[K]/[CO2]**
4. **Monoamine oxidase (MAO**) is an enzyme which breaks catecholamines. (Give marks if attempted). **[U]/[CO2]**
5. The threshold for triggering an action potential is typically around **-55** mV. **[K]/[CO2]**
6. Sensory neurons (or afferent neurons) carry impulses from sensory receptors toward the **Central nervous system/brain.** **[K]/[CO2]**
7. **Dopamine** is the primary neurotransmitter involved in forebrain activity linked to reward mechanisms. **[U]/[CO3,CO4]**
8. The hypothalamus communicates with the autonomic nervous system to regulate involuntary functions such as **(Any involuntary actions like digestion, respiration, blood pressure, temperature control etc).** **[U]/[CO3,CO4]**
9. The left hemisphere of the brain is primarily responsible for **language/analytical reasoning** processing. **[K]/[CO3,CO4]**
10. The **medulla oblongata** houses the cardiac, respiratory, and vasomotor control centers. **[K]/[CO3,CO4]**

Part B

Short Questions
Answer any 10 Questions
Each Question carries 2 marks

11. Write about the bodily factors that influence the mind. [U]/[CO1]

- Hormones, Neurotransmitters, Nutrition, Sleep cycles, Physical health. (Mention any 2 factors or a brief account of any 1 factors). (Liberal Valuation.)

12. Define BMR. [K]/[CO1]

- BMR (Basal Metabolic Rate) is the minimum amount of energy required by the body to maintain vital physiological functions—such as breathing, circulation, cell production, and maintaining body temperature—while at complete physical and mental rest, in a post-absorptive state. (full form- 0.5 mark, definition : 1.5 mark) **(Liberal Valuation. Give full marks if attempted)**

13. What are ependymal cells and explain its role in nervous system? [U]/[CO2]

- Ependymal cells are macroglial cells found lining the ventricles of the brain and the central canal of the spinal cord, and they help in producing and circulating cerebrospinal fluid (CSF). (definition :1 mark , function :1 mark)

14. What is the significance of K+ leak channels in resting potential? [U]/[CO2]

- K^+ leak channels maintain the resting potential by allowing K^+ to exit the cell, keeping the inside negative.

15. Explain the role of Ranvier's nodes in impulse transmission. [U]/[CO2]

- Nodes of Ranvier are gaps in the myelin sheath where the axonal membrane is exposed.
- Nodes of Ranvier speed up nerve impulses by allowing saltatory conduction. (give full marks for a good account)

16. Differentiate between short-term and long-term potentiation. [U]/[CO2]

- Short-Term Potentiation (STP):
- A brief increase in synaptic strength lasting seconds to minutes, mainly due to increased neurotransmitter release.
- Long-Term Potentiation (LTP):
- A long-lasting enhancement of synaptic strength lasting hours to years, involving protein synthesis and formation of new receptors/synapses, important for memory. (Liberal valuation)

17. Comment on the types of neuroglial cells. [U]/[CO2]

- Neuroglial cells support, protect, and nourish neurons. The major types are:
- **Astrocytes:** Provide structural support, regulate nutrients, maintain the blood–brain barrier.
- **Oligodendrocytes:** Form myelin sheath in the CNS..
- **Ependymal cells:** Line ventricles and central canal; produce and circulate CSF.
- **Microglia:** Act as immune cells; remove debris and pathogens. (0.5 marks each)

18. **How does the parasympathetic nervous system help maintain the homeostasis ? [U]/[CO3,CO4]**

The parasympathetic nervous system maintains homeostasis by restoring the body to a calm, resting state. It:

- Slows heart rate and lowers blood pressure, conserving energy.
- Stimulates digestion, gland secretion, and nutrient absorption, supporting normal body functions.
- Promotes relaxation and recovery after stress. (give full marks for a good account).

19. **Describe the physiologic changes happening during stimulation of sympathetic fibres. [U]/[CO3,CO4]**

Stimulation of sympathetic fibres produces the **"fight-or-flight" response**, causing:

- Increased heart rate and blood pressure
- Bronchodilation (more airflow to lungs)
- Pupil dilation
- Increased blood glucose due to glycogen breakdown
- Reduced digestion (decreased gut motility and secretions)
- Increased sweating and alertness (Mention flight or fight response and any three responses, 0.5 marks each)

20. **Describe the role of the cranial nerves in the peripheral nervous system. [U]/[CO3,CO4]**

- Control sensory and motor functions of the head and neck, including vision, hearing, smell, taste, facial movements, and swallowing.
- Provide parasympathetic outflow to organs such as the heart, lungs, and digestive system (mainly via the vagus nerve), helping regulate involuntary functions.
- They are essential for communication between the brain and structures of the body outside the spinal cord.(Any two function: 1 mark each)

21. **What are the functions of the brainstem in the central nervous system? [An]/[CO3,CO4]**

- Regulating essential life functions such as breathing, heart rate, and blood pressure.
- Acting as a relay center, connecting the brain and spinal cord for sensory and motor pathways.
- Controlling reflexes like swallowing, coughing, and vomiting.
- Maintaining consciousness and sleep–wake cycles through the reticular formation. (Mention any two function)

22. **Describe the structure and function of the meninges in brain. [U]/[CO3,CO4]**

The meninges are three protective layers surrounding the brain:

- Dura mater – the tough, outermost layer.
- Arachnoid mater – the middle, web-like layer.
- Pia mater – the thin, delicate inner layer that closely covers the brain surface.

Function:

- Protect the brain from injury.
- Enclose and support blood vessels.
- Contain cerebrospinal fluid (CSF) in the subarachnoid space, which cushions the brain and maintains chemical stability. (Explanation of structures: 1 mark, Any two function: 0.5 mark each)

Part C

Short Essay Type Questions
Answer any 5 Questions
Each Question carries 4 marks

23. Write a short note on different stages of human evolution. [K]/[CO1]

- Human evolution occurred through several stages showing increases in brain size, upright posture, and tool use:
- Australopithecus: Early bipedal ancestors with small brains.
- Homo habilis: First tool users ("handy man").
- Homo erectus: Fully upright, used fire, migrated out of Africa.
- Neanderthals: Strong, intelligent humans with advanced tools.
- Homo sapiens: Modern humans with complex language, culture, and high intelligence. (Give full marks for a good account).

24. Describe the legal system that regulates animal research in the U.S. and the role of the Institutional Animal Care and Use Committee. [U]/[CO1]

- Animal research in the U.S. is regulated mainly by the Animal Welfare Act (AWA) and Public Health Service (PHS) Policy, which ensure humane treatment, proper housing, and ethical use of animals.
- The Institutional Animal Care and Use Committee (IACUC) reviews and approves all animal research, ensures the 3Rs (Replacement, Reduction, Refinement), inspects facilities, and monitors compliance with federal laws. (Liberal valuation)

25. Describe the stages of action potential with the help of a labeled diagram. [U]/[CO2]

1. Resting Potential:
   Neuron is at –70 mV, maintained by K^+ leak channels and Na^+ /K^+ pump.
2. Depolarization:
   Threshold is reached; voltage-gated Na^+ channels open, Na^+ rushes in, and the membrane potential becomes positive.
3. Repolarization:
   Na^+ channels close; voltage-gated K^+ channels open, K^+ moves out, restoring negativity.
4. Hyperpolarization:
   K^+ channels close slowly causing slight overshoot; membrane returns to resting level.(Explanation of all stages: 3 mark, diagram : 1 mark).

26. Discuss excitatory post synaptic potential. [K]/[CO2]

- An EPSP is a small depolarization in the postsynaptic neuron that makes it more likely to fire an action potential.
- It happens when excitatory neurotransmitters open Na^+ or Ca^{2+} channels, allowing positive ions to enter and making the inside of the neuron less negative. (Give full marks for a good account)

27. How does the midbrain contribute to the regulation of movement and coordination? [An]/[CO3,CO4]

- The midbrain helps regulate movement and coordination by acting as a control center that connects motor signals between the brain and spinal cord. It contains important

structures like the substantia nigra, which produces dopamine needed for smooth and purposeful movements, and the red nucleus, which helps coordinate limb movements. The midbrain also integrates sensory information—especially visual and auditory cues—to guide reflexive actions and ensure the body adjusts its movements accurately.

28. Explain the role of the spinal cord in reflex action. [U]/[CO3,CO4]

- The spinal cord acts as the reflex center of the body.
- Reflex actions are automatic, rapid, and involuntary responses to stimuli.
- Sensory neurons carry signals from receptors to the spinal cord.
- The spinal cord processes the information without sending it to the brain.
- Motor neurons then carry the response from the spinal cord to the muscles or glands.
- This quick pathway is called a reflex arc.
- It helps protect the body by allowing immediate responses to danger (e.g., withdrawing hand from fire). (Give full marks for a good account)

29. Give an account on the structure of neocortex. [U]/[CO3,CO4]

The neocortex is the six-layered outer region of the cerebral cortex responsible for higher functions like perception, learning, and decision-making. Its layers (I–VI) are:

1. Molecular Layer (Layer I) – Few neurons; mainly nerve fibers and synapses.
2. External Granular Layer (Layer II) – Small granular and pyramidal neurons.
3. External Pyramidal Layer (Layer III) – Medium pyramidal neurons; cortical connections.
4. Internal Granular Layer (Layer IV) – Densely packed granule cells; receives sensory input from the thalamus.
5. Internal Pyramidal Layer (Layer V) – Large pyramidal cells; major motor output to brainstem and spinal cord.
6. Multiform Layer (Layer VI) – Mixed cell types; sends signals back to the thalamus.

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